

```

> restart;
with(student) :
with(linalg) :

# =====
#      4
#      1
#
#
# =====

```

```

> assume(g > 0, n :: posint);

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>
pde := diff(u(x, t), t) = diff(u(x, t), x$2) - u(x, t);
init := u(x, 0) = 1;
bound := u(0, t) = 0, u(g, t) = 0;

```

$$pde := \frac{\partial}{\partial t} u(x, t) = \frac{\partial^2}{\partial x^2} u(x, t) - u(x, t)$$

$$init := u(x, 0) = 1$$

$$bound := u(0, t) = 0, u(g, t) = 0$$

(1)

```

>
#
subs(u(x, t) = X(x) * T(t), pde) :
expand(lhs(%) / (X(x) * T(t))) = expand(rhs(%) / (X(x) * T(t)));

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$$\frac{\frac{d}{dt} T(t)}{T(t)} = \frac{\frac{d^2}{dx^2} X(x)}{X(x)} - 1$$

(2)

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>
lambda_eq_T := diff(T(t), t) + lambda * T(t) = 0;
lambda_eq_X := diff(X(x), x$2) + (lambda - 1) * X(x) = 0;

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$$\text{lambda_eq_T} := \frac{d}{dt} T(t) + \lambda T(t) = 0$$

$$\text{lambda_eq_X} := \frac{d^2}{dx^2} X(x) + (\lambda - 1) X(x) = 0 \quad (3)$$

>

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#
solX := dsolve(lambda_eq_X, X(x));
X := unapply(rhs(solX), x);
e1 := X(0) = 0;
e2 := X(g) = 0;
```

$$\text{solX} := X(x) = c_1 \sin(\sqrt{\lambda - 1} x) + c_2 \cos(\sqrt{\lambda - 1} x)$$

$$X := x \quad c_1 \cdot \sin(\sqrt{\lambda - 1} \cdot x) + c_2 \cdot \cos(\sqrt{\lambda - 1} \cdot x)$$

$$e1 := c_2 = 0$$

$$e2 := c_1 \sin(\sqrt{\lambda - 1} g) + c_2 \cos(\sqrt{\lambda - 1} g) = 0 \quad (4)$$

>

```
_EnvAllSolutions := true;
solve(sin(sqrt(lambda - 1) * g) = 0, lambda);
subs(_Z1 = n, %);
ev := unapply(%, n);
```

$$_EnvAllSolutions := true$$

$$\frac{\pi^2 Z1^2 + g^2}{g^2}$$

$$\frac{\pi^2 n^2 + g^2}{g^2}$$

$$ev := n \sim \frac{\pi^2 \cdot n^2 + g^2}{g^2} \quad (5)$$

>

```
# ( )
phi := (n, x) sqrt(2/g) * sin(n * Pi * x / g);
```

$$\phi := (n, x) \quad \sqrt{\frac{2}{g}} \cdot \sin\left(\frac{n \cdot \pi \cdot x}{g}\right) \quad (6)$$

>

```
#
T_n := t exp(-ev(n) * t);
```

$$T_n := t \quad e^{-ev(n) \cdot t} \quad (7)$$

>

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# ,
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A_n := Int(1 * phi(n, x), x = 0 .. g);
A_n := value(A_n);
A_n := simplify(A_n) assuming n :: posint, g > 0;
A_n := subs(sin(n * Pi) = 0, cos(n * Pi) = (-1)^n, A_n);
A_n := simplify(A_n);

```

$$A_n := \int_0^g \frac{\sqrt{2} \sin\left(\frac{n\pi x}{g}\right)}{\sqrt{g}} dx$$

$$A_n := -\frac{\sqrt{2} \sqrt{g} ((-1)^n - 1)}{n\pi}$$

$$A_n := -\frac{\sqrt{2} \sqrt{g} ((-1)^n - 1)}{n\pi}$$

$$A_n := -\frac{\sqrt{2} \sqrt{g} ((-1)^n - 1)}{n\pi}$$

$$A_n := -\frac{\sqrt{2} \sqrt{g} ((-1)^n - 1)}{n\pi} \quad (8)$$

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>
#
term := A_n * T_n(t) * phi(n, x);

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$$term := -\frac{2((-1)^n - 1) e^{-\frac{(\pi^2 n^2 + g^2)t}{g^2}} \sin\left(\frac{n\pi x}{g}\right)}{n\pi} \quad (9)$$

```

> u_series := Sum(term, n = 1 .. infinity);

```

$$u_series := \sum_{n=1}^{\infty} \left(-\frac{2((-1)^n - 1) e^{-\frac{(\pi^2 n^2 + g^2)t}{g^2}} \sin\left(\frac{n\pi x}{g}\right)}{n\pi} \right) \quad (10)$$

```

>
#
printf(" '      ( 4 . 3 ,      1 ) : \ n "
      );
u_series;
'      ( 4 . 3 ,      1 ) :

```

$$\sum_{n=1}^{\infty} \left(-\frac{2((-1)^n - 1) e^{-\frac{(\pi^2 n^2 + g^2)t}{g^2}} \sin\left(\frac{n\pi x}{g}\right)}{n\pi} \right) \quad (11)$$

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> # =====
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# 1.      :   t = 0      u ( x , 0 ) = 1
#      ( , 1 0 0 )   g = 1 ,   t = 0
g_val := 1 :
N_terms := 100 :
u_approx := unapply( sum( eval( term, [g = g_val, t = 0] ), n = 1 .. N_terms ), x ) :
printf( "\n      u ( x , 0 ) = 1      g = 1 : \n " );
printf( " u ( 0 . 5 , 0 ) : %f ( 1 ) \n " , u_approx(0.5) );
printf( " u ( 0 . 2 , 0 ) : %f ( 1 ) \n " , u_approx(0.2) );
printf( " u ( 0 . 8 , 0 ) : %f ( 1 ) \n \n " , u_approx(0.8) );
```

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      u ( x , 0 ) = 1      g = 1 :
u ( 0 . 5 , 0 )      :   0 . 9 9 3 6 3 4   ( 1 )
u ( 0 . 2 , 0 )      :   0 . 9 8 9 1 7 4   ( 1 )
u ( 0 . 8 , 0 )      :   0 . 9 8 9 1 7 4   ( 1 )
```

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# 2.      :   u ( 0 , t ) = 0   u ( g , t ) = 0
#   x = 0   x = g   s i n ( . . . ) = 0 ,   u = 0 .
printf( "      : \n " );
printf( "u(0,t) = %s (sin(0)=0) => \n " , "0");
printf( "u(g,t) = %s (sin(nπ)=0) => \n \n " , "0");

:
u ( 0 , t ) = 0   ( s i n ( 0 ) = 0 ) =>
u ( g , t ) = 0   ( s i n ( n & pi ; ) = 0 ) =>
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```
# 3.      ,      ( )
#   n = 1 ,      P D E
n_test := 1 :
term_test := eval( term, n = n_test ) :
#   u _ t   u _ x x
ut := diff( term_test, t ) :
uxx := diff( term_test, x$2 ) :
#   u t = u x x - u
residual := simplify( ut - ( uxx - term_test ) );
printf( "      ( n = 1 ) : \n " );
printf( " = %a = 0 = > \n \n " , residual );
```

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residual := 0
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      ( n = 1 ) :
= 0 = 0 =>

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# 4.      ( g = 1 , t = 0 .. 0.5 )
g_val := 1 :
u_plot := unapply( sum( eval( term, g = g_val ), n = 1 .. 50 ), x, t ) :
plot3d( u_plot( x, t ), x = 0 .. 1, t = 0 .. 0.5, axes = boxed, title = "    u ( x , t ) " );

```

u (x , t)



